GCI FINAL PROJECT

ENHANCING EMPLOYMENT SOLUTION WITH DATA ANALYTICS

Recent Consumer Industries

Many industries anguished high rate of employee attrition due to limited development of industrial plan, which can lead some of industries bankruptcy. As a result, the goals may not accomplish which are desired. Based on that, we set several problems to carry out a solution of the business plan.

- Lack of advancement opportunities which can lead over 20% of employees would be 'somewhat' likely to leave their job in company which is limited opportunities for development.
- Poor work life balance and physical work environments can drive employees away, such as outdated facilities, dissatisfaction between manager and employee.
- Poor leadership and management can significantly lead high attrition due to ineffective communication, lack of support.

Main References:

"6 Reasons Behind Employee Attrition (and how to fix them!)" (Instep) https://instepuk.com/article/employee-attrition/

"Employee Attrition: The Definitive Guide" (Primalogik) https://primalogik.com/blog/employee-attrition-definitive-guide/#:~:text=As%20a%20result%2C%20the%20team,are%20leaving%20without%20being%20replaced.

"IBM HR Analytics Employee Attrition & Performance" (Kaggle) https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Introduction of Data

Approximately 1470 types of customer information have been accumulated as data as shown in below.

	Age	Attrition	BusinessTravel	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisf
0	41	Yes	Travel_Rarely	Sales	1	2	Life Sciences	1	1	
1	49	No	Travel_Frequently	Research & Development	8	1	Life Sciences	1	2	
2	37	Yes	Travel_Rarely	Research & Development	2	2	Other	1	4	
3	33	No	Travel_Frequently	Research & Development	3	4	Life Sciences	1	5	
4	27	No	Travel_Rarely	Research & Development	2	1	Medical	1	7	

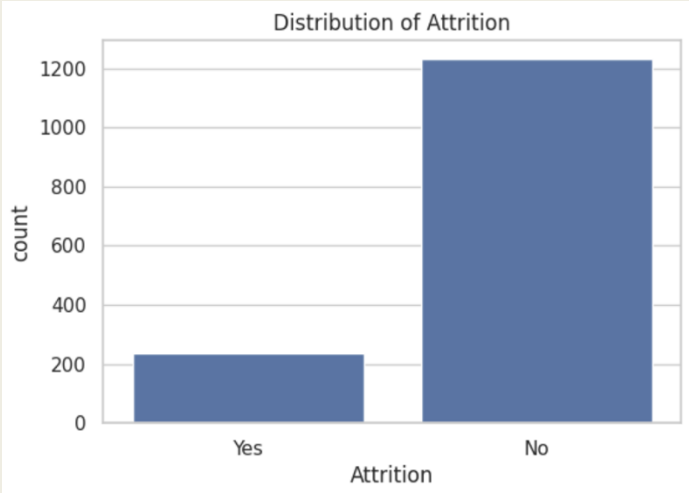
A portion of the data you provided

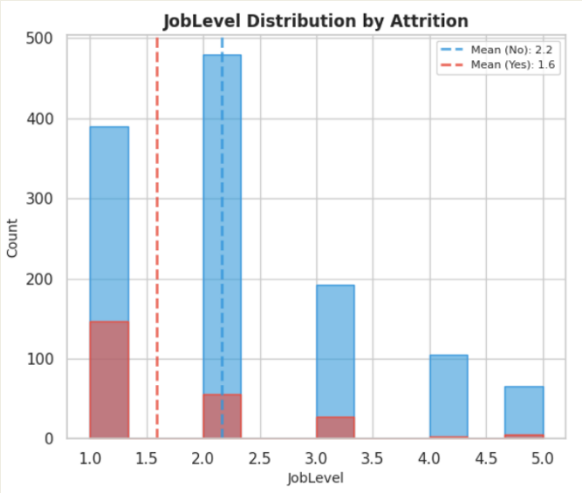
Data Contents:

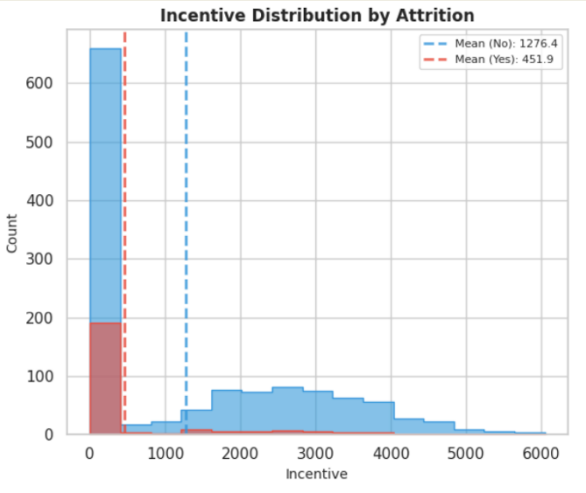
- Key data for each employee (Employee Number, Gender, Job Role, Age, Marital Status, Business Travel, Department, Education, Distance from Home, Years at Company and Monthly Income)
- Information on employees' past year performance (Monthly Achievement, Performance Index, Over Time, etc.)
- Incentive on employees
- Frequency Training Last Year
- Years with involvement

Data Visualization

We conducted various visualizations of employee data. But we display some key visualizations below.





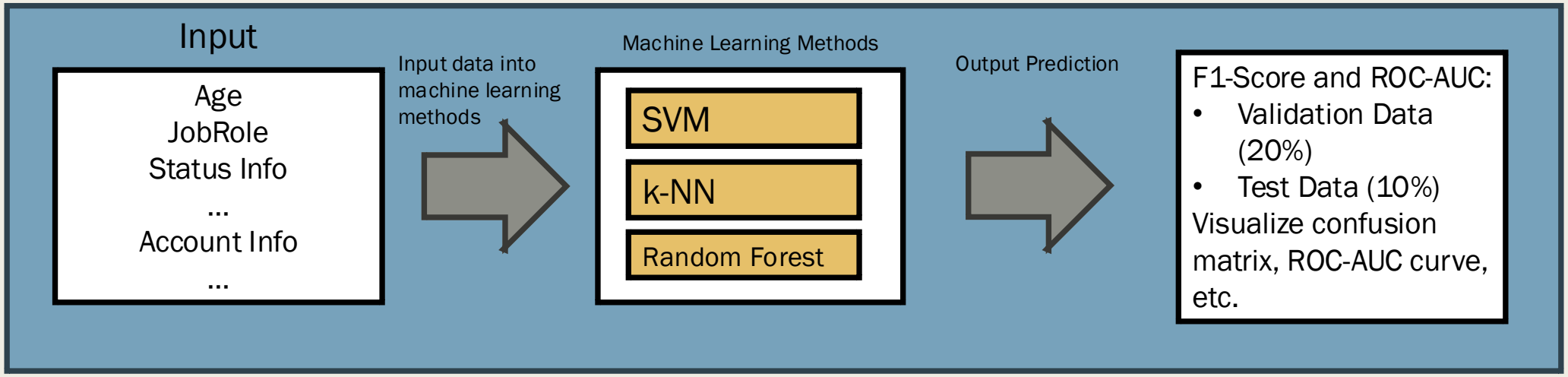


- Around 16% employee attrition suffered in this company.
- High attrition mostly from junior employees.
- Inadequate incentives are likely driving employees away, especially every career-staff.

Difficult to identify what are factors which can lead high attrition. The solution is by using machine learning methods

Overview of Machine Learning Model

From this employee information, we proposed three machine learning models as a below.

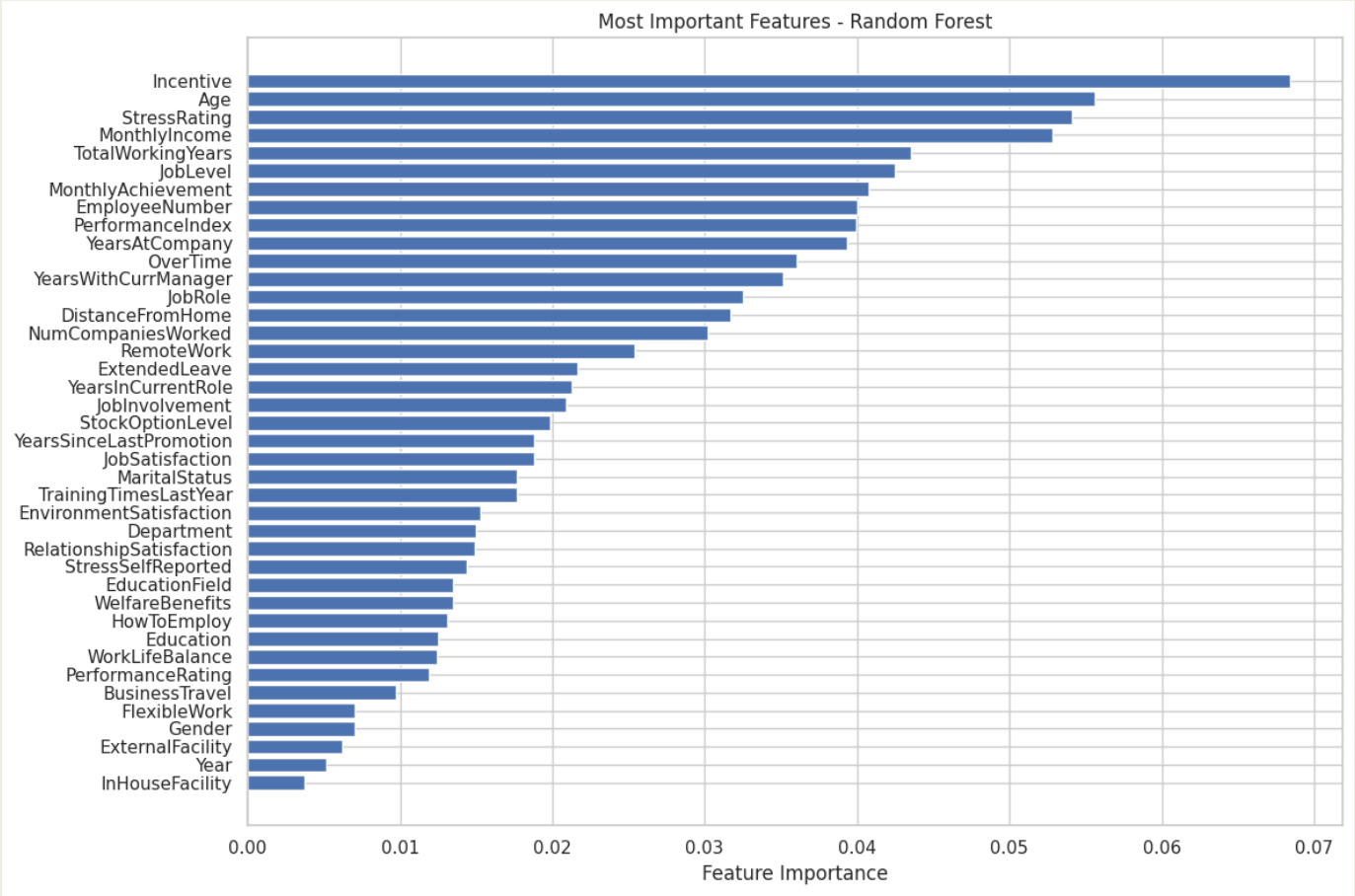


Since the target value is imbalanced, we resample this by using resample package which balance between the 2 target value so that it make better accuracy for predicting.

Importance Features for Prediction

Many of the features important for prediction are those related to effect of attrition.

By utilizing both AI and human decision-making with key features, led a high accurate lending decisions become possible.



OUR BUSINESS PROPOSAL

Our Business Proposal

Our goal for business proposal is to predict the circumstances of employment rate in industries against attrition in long-term plan.

These accommodates:

- Estimation of attrition rate in long-term future with machine learning methods
- Through various analyses and training, further accuracy improvement is possible like other metrics.

Presupposition Attrition Rate in Long-term Plan

By utilizing AI, fast, and validity of machine learning models become possible.

Overview

Using resampled dataset, predict rate of attrition in each company with reliable machine learning.

Novelty

Decisions that were traditionally made based on job level and total employees' working year.

Effects

- Employee: How satisfies and performance by manager
- Company: More employees leading new generation for long-term plan

Points

Ideally, we would like to analyze how the attrition employee rate goes in the future based on the reliable AI.

Reliable Machine Learning

Using various machine learning methods, we got the metrics score below.

	Val-F1	Test-F1	Val-AUC	Test-AUC
SVM	0.682	0.669	0.710	0.677
k-NN	0.606	0.625	0.700	0.696
Random Forest	0.977	0.987	0.999	0.999

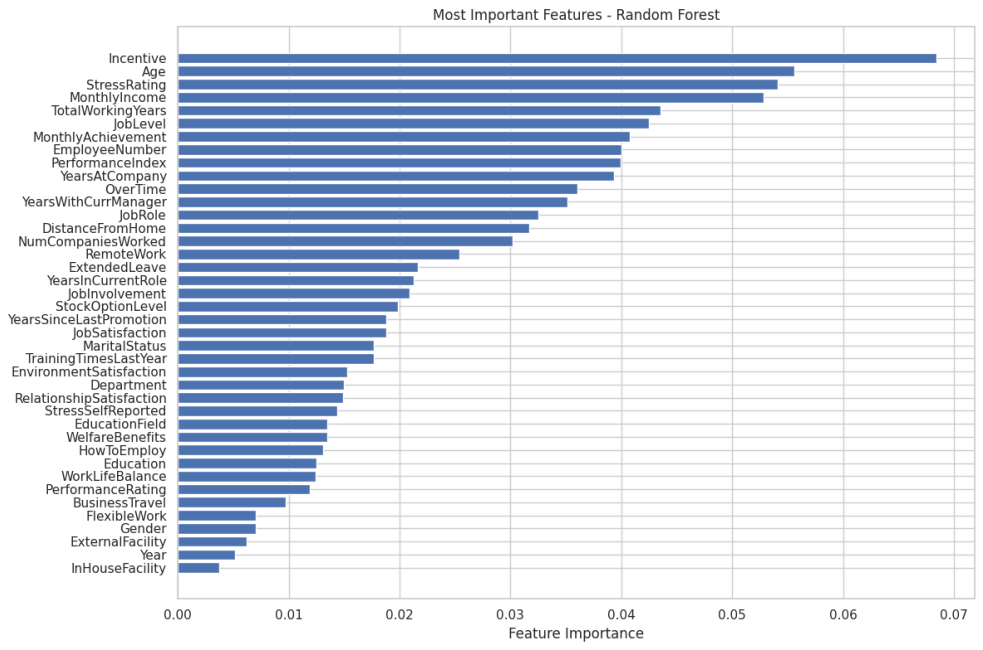
From this table on the left, Random Forest has the best model, but SVM model has more reliable than other two models to avoid overfitting. So we use SVM as a valid model

Number of Validation Data = 484		Prediction	
		Yes	No
Results	Yes	165	77
	No	77	165

Among all employees 68% can be detected. On the other hand 77 employees (32% of employees with attrition) were mistakenly predicted. Therefore, the classification of model is balanced.

We may conclude that 50% employees will go away as predicted for long-term, or it increased by 34% in look-out circumstances.

Impact on Attrition Rate



Features are the most important in impact on attrition employee rate

Factors that affected most:

- Inadequate incentive
- Ageing employees’
- Employees’ stress level
- Monthly Income isn’t in accordance with

More than 5% were affected by 4 factors common for the problem employment company.

Avoid on employment attrition

With awareness of increasing unemployment rate, we will guide how to avoid any problems!

- Offer new training and development programmes for managers and owner to help gain leadership skills for next plan.
- Invest new improved physical work environment to attract employees.
- To ensure that salaries and benefits are competitive and fair, try to conduct regular market research to stay updated on industry standards and adjust compensation packages accordingly.

Quotation

Content

- According to the predictions, the unemployment rate will increased by around 34%.
- The common factors for the impact of attrition rate was incentive, age employee, stress rating and monthly income for each customer.

Price

We proposed a fee of 3 million dollar for all companies for improving junior employees and accelerate business long-term plans.